```
C:\Users\Admin\PycharmProjects\PythonProject2\.venv\Scripts\python.exe "C:\Users\Admin\PycharmProjects\PythonProject2\Data Selection.py"
Dataset Loaded

First 5 rows
    Name  Age Department  Marks       City
0   Asha   20        CSE   65.0       Pune
1  Meena   24      Civil   75.0  Hyderabad
2   Isha   21         IT   78.0       Pune
3  Varun   24        CSE   82.0       Pune
4  Varun   21         IT   76.0      Delhi

Last 5 rows
       Name  Age Department  Marks       City
115    Ravi   24         IT   61.0     Mumbai
116   Sneha   23         IT   77.0  Bangalore
117  Vikram   20        CSE   72.0  Bangalore
118    Isha   21         ME   91.0       Pune
119   Priya   24         ME   80.0      Delhi

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Name        120 non-null    str
 1   Age         120 non-null    int64
 2   Department  120 non-null    str
 3   Marks       118 non-null    float64
 4   City        120 non-null    str
dtypes: float64(1), int64(1), str(3)
memory usage: 4.8 KB
None

Statistical Summary
              Age       Marks
count  120.000000  118.000000
mean    22.791667   79.135593
std      1.603021   11.555822
```

```
min     20.000000   60.000000
25%     22.000000   69.250000
50%     23.000000   79.000000
75%     24.000000   88.000000
max     25.000000   99.000000

Select single column (Marks)
0      65.0
1      75.0
2      78.0
3      82.0
4      76.0
       ...
115    61.0
116    77.0
117    72.0
118    91.0
119    80.0
Name: Marks, Length: 120, dtype: float64

Select multiple columns (Name, Marks)
       Name  Marks
0      Asha   65.0
1     Meena   75.0
2      Isha   78.0
3     Varun   82.0
4     Varun   76.0
..      ...    ...
115    Ravi   61.0
116   Sneha   77.0
117  Vikram   72.0
118    Isha   91.0
119   Priya   80.0

[120 rows x 2 columns]

First 5 rows using iloc
     Name  Age Department  Marks       City
0   Asha   20        CSE   65.0       Pune
1  Meena   24      Civil   75.0  Hyderabad
2   Isha   21         IT   78.0       Pune
```

```
 3  Varun   24        CSE   82.0       Pune
 4  Varun   21         IT   76.0      Delhi

Select rows 0-5 and columns 0-2
     Name  Age Department
0   Asha   20        CSE
1  Meena   24      Civil
2   Isha   21         IT
3  Varun   24        CSE
4  Varun   21         IT

Select rows and specific columns
      Name  Marks Department
0    Asha   65.0        CSE
1   Meena   75.0      Civil
2    Isha   78.0         IT
3   Varun   82.0        CSE
4   Varun   76.0         IT
5   Pooja   61.0      Civil
6   Meena   63.0      Civil
7    Ravi   72.0         IT
8   Divya   79.0        CSE
9   Pooja   68.0         IT
10  Rahul   71.0         IT

Students with Marks > 85
        Name  Age Department  Marks       City
13      Ravi   21         IT   99.0  Bangalore
14     Arjun   21      Civil   98.0    Chennai
16     Pooja   21        ECE   87.0  Hyderabad
23     Priya   22        CSE   94.0     Mysore
27      Ravi   20        CSE   95.0     Mysore
28     Pooja   25        ECE   88.0     Mumbai
33     Anita   23        ECE   94.0     Mysore
35     Arjun   22      Civil   98.0    Chennai
39     Rahul   22        ECE   91.0    Chennai
41    Nikhil   24        ECE   96.0       Pune
47      Ravi   24         IT   88.0      Delhi
48     Kavya   25         ME   92.0       Pune
51     Varun   21      Civil   92.0     Mumbai
52     Meena   20      Civil   91.0     Mysore
```

```
122 53     Rahul   23       Civil   86.0       Delhi
123 54     Kiran   25          ME   93.0      Mumbai
124 56      Asha   20          IT   93.0      Mysore
125 57     Meena   22          ME   89.0   Hyderabad
126 59     Arjun   24          IT   96.0   Hyderabad
127 60     Rohit   25         CSE   94.0      Mysore
128 64     Kiran   23          ME   93.0   Bangalore
129 70     Sneha   21         CSE   97.0       Pune
130 71      Ravi   21         CSE   93.0     Chennai
131 72     Varun   22          IT   86.0      Mumbai
132 80    Nikhil   24       Civil   99.0   Hyderabad
133 81     Rahul   23         ECE   88.0     Chennai
134 83     Kavya   24          IT   93.0      Mumbai
135 89     Kiran   25          IT   96.0   Bangalore
136 94     Arjun   20          IT   98.0       Delhi
137 100    Kavya   24         CSE   86.0       Delhi
138 101     Ravi   22         CSE   89.0       Pune
139 102     Ravi   21         CSE   97.0   Hyderabad
140 103    Rohit   23         CSE   86.0      Mumbai
141 107    Manoj   21          IT   99.0       Pune
142 109    Varun   24          ME   97.0   Hyderabad
143 114    Priya   22          ME   94.0      Mysore
144 118     Isha   21          ME   91.0       Pune
145
146 Students from CSE Department
147         Name  Age Department  Marks       City
148 0       Asha   20         CSE   65.0       Pune
149 3      Varun   24         CSE   82.0       Pune
150 8      Divya   25         CSE   79.0       Delhi
151 18    Nikhil   23         CSE   76.0   Hyderabad
152 19      Amit   22         CSE   82.0   Bangalore
153 23     Priya   22         CSE   94.0      Mysore
154 26     Varun   24         CSE   83.0       Delhi
155 27      Ravi   20         CSE   95.0      Mysore
156 38     Anita   23         CSE   85.0       Pune
157 42     Priya   23         CSE   70.0   Bangalore
158 50     Meena   20         CSE   66.0       Pune
159 58     Rohit   25         CSE   69.0     Chennai
160 60     Rohit   25         CSE   94.0      Mysore
161 66      Amit   25         CSE   77.0       Pune
162 67     Pooja   21         CSE   79.0       Pune
```

```
69    Rahul   22        CSE   64.0     Mumbai
70    Sneha   21        CSE   97.0       Pune
71     Ravi   21        CSE   93.0    Chennai
91    Manoj   21        CSE   79.0      Delhi
100   Kavya   24        CSE   86.0      Delhi
101    Ravi   22        CSE   89.0       Pune
102    Ravi   21        CSE   97.0  Hyderabad
103   Rohit   23        CSE   86.0     Mumbai
110   Pooja   25        CSE   65.0     Mysore
117  Vikram   20        CSE   72.0  Bangalore

CSE students with Marks > 85
      Name  Age Department  Marks       City
23   Priya   22        CSE   94.0     Mysore
27    Ravi   20        CSE   95.0     Mysore
60   Rohit   25        CSE   94.0     Mysore
70   Sneha   21        CSE   97.0       Pune
71    Ravi   21        CSE   93.0    Chennai
100  Kavya   24        CSE   86.0      Delhi
101   Ravi   22        CSE   89.0       Pune
102   Ravi   21        CSE   97.0  Hyderabad
103  Rohit   23        CSE   86.0     Mumbai

Sorted by Marks (Descending)
       Name  Age Department  Marks       City
13     Ravi   21         IT   99.0  Bangalore
80   Nikhil   24      Civil   99.0  Hyderabad
107   Manoj   21         IT   99.0       Pune
94    Arjun   20         IT   98.0      Delhi
35    Arjun   22      Civil   98.0    Chennai
..      ...  ...        ...    ...        ...
65    Meena   25      Civil   60.0     Mumbai
86     Ravi   22         ME   60.0     Mysore
112  Vikram   22         ME   60.0     Mumbai
11    Meena   22         ME    NaN      Delhi
29    Rohit   25         IT    NaN     Mysore

[120 rows x 5 columns]

Missing Values Count
Name          0
```

```
Age           0
Department    0
Marks         2
City          0
dtype: int64

Dataset after filling missing values
     Name  Age Department  Marks       City
0   Asha   20        CSE   65.0       Pune
1  Meena   24      Civil   75.0  Hyderabad
2   Isha   21         IT   78.0       Pune
3  Varun   24        CSE   82.0       Pune
4  Varun   21         IT   76.0      Delhi

Average Marks by Department
Department
CSE      80.960000
Civil    78.411765
ECE      77.000000
IT       78.382043
ME       81.322926
Name: Marks, dtype: float64

Department Statistics
                Marks                    Age
                 mean   max   min       mean
Department
CSE         80.960000  97.0  64.0  22.480000
Civil       78.411765  99.0  60.0  22.705882
ECE         77.000000  96.0  61.0  22.863636
IT          78.382043  99.0  61.0  22.972973
ME          81.322926  97.0  60.0  22.842105

Process finished with exit code 0
```